

EPOROD

Production of Fiber reinforcement Polymer

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Introduction

Shahkar Fekr Tavana Company, with registration number 5296, is the founder of the production of FRP nanocomposite rebars with epoxy resin in Iran, which has always put innovation in the production of products required by a wide range of customers at the forefront of its work relying on the vast science of nanopolymers and nanocomposites as well as technical, engineering and structural knowledge and information and the use of advanced machines and serious monitoring of product quality, and has been able to prevent capital outflow and dependence on the countries that produce this product and reduce the cost of the product drastically. The experienced founders of this company started their research on GFRP rebars in 2014 and finally in 2015 they succeeded in localizing production lines and also producing GFRP rebars on an industrial scale for the first time in Iran with international standards. Using new and innovative technologies, these products have been able to be considered as a suitable and efficient replacement for steel bars.

In addition to obtaining the necessary permits, this company was able to obtain the certificate of membership in the Iranian Composite Association and the registration of the exclusive and international EPOROD brand, as well as obtaining the standard certificates of CE (product export mark), IMS (Integrated Management System) and HSE-MS (Health, Safety and Environment Management System) and ISO9001:2015, ISO 14001:2015 and ISO 45001:2018 from Canada.

EPOROD's production lines currently include production lines for composite bends and stirrup, composite mesh, ribbed and non-ribbed composite rebar, and anchor-head

All these production lines have been designed and built by EPOROD company's equipment and production line team using advanced equipment and modern technologies. Also, the elite team of EPOROD company has always sought to improve and optimize the performance of its production lines by using its technical and experimental knowledge.



Overall Structure of EPOROD

Technical Engineering Company

EPOROD Technical Engineering Company currently consists of 5 strong and expert groups, each of which has a professional and complete structure, that has finally led to the building of production lines and products with international standards and quality. Some of these structures are mentioned below:

1) R & D or Research and Development group: which consists of two subgroups: Group 1) Research and development group of chemical, polymer and composite materials and products – AND , Group 2) Research and development group of production lines and new industrial equipment.

2) Sales and Marketing group: which includes more than 100 active sales agents inside the country and more than 15 international sales agents for abroad sales

3) Production group: This group consists of 25 people with expertise and experience in the fields of production, electrical engineering, mechanical engineering, polymer engineering, production managers and assembly workers who are based in the city of Dezful.

4) Administrative group and Managers: Include people with experience and expertise in various fields, including commercial manager, financial manager, factory manager, CEO, sales manager, manager of the research and development group, manager of the construction and equipment group, human resources manager, manager of legal affairs and contracts, etc., who are also located in the cities of Neyshabur and Dezful.

5) Equipment and Production Line Manufacturing group: which includes 12 experts and experienced people in the fields of industrial drawing and design, and operators and master craftsmen with experience in industrial machines (such as lathes, wire cut, CNC lathes, mills, etc.) and industrial electric engineers, mechanical engineers and etc. This group is responsible for the task of designing and equipping production lines and manufacturing production machines, as well as providing after-sales services and technical support in relation to all kinds of EPOROD production lines. This production line construction complex is located in Neyshabur city in a roofed and complete area with more than 1000 square meters.

EPOROD BEYOND THE EXPECTATIONS

A View of Equipment and Machinery for EPOROD Group Production Lines











why composite rebar?

As you know, concrete is inherently strong enough to bear compressive forces, but tensile forces cause concrete to break. Rebars are used in concrete to strengthen concrete and heavy structures in bearing tensile forces, which are called final reinforced concrete structure. Since 1968, steel bars have been used to increase the tensile strength of concrete.

Despite their unique functional characteristics, steel rebars are severely exposed to harsh environmental conditions, i.e. high humidity, very cold or hot environments, alkaline and acidic conditions, salts in the air, etc. and hence they are highly prone to damages caused by corrosion which reduce the life of structures and destroy them.

Another major problem of steel bars is their high weight compared to their strength. The high weight of the rebars causes dead loads in the structure.

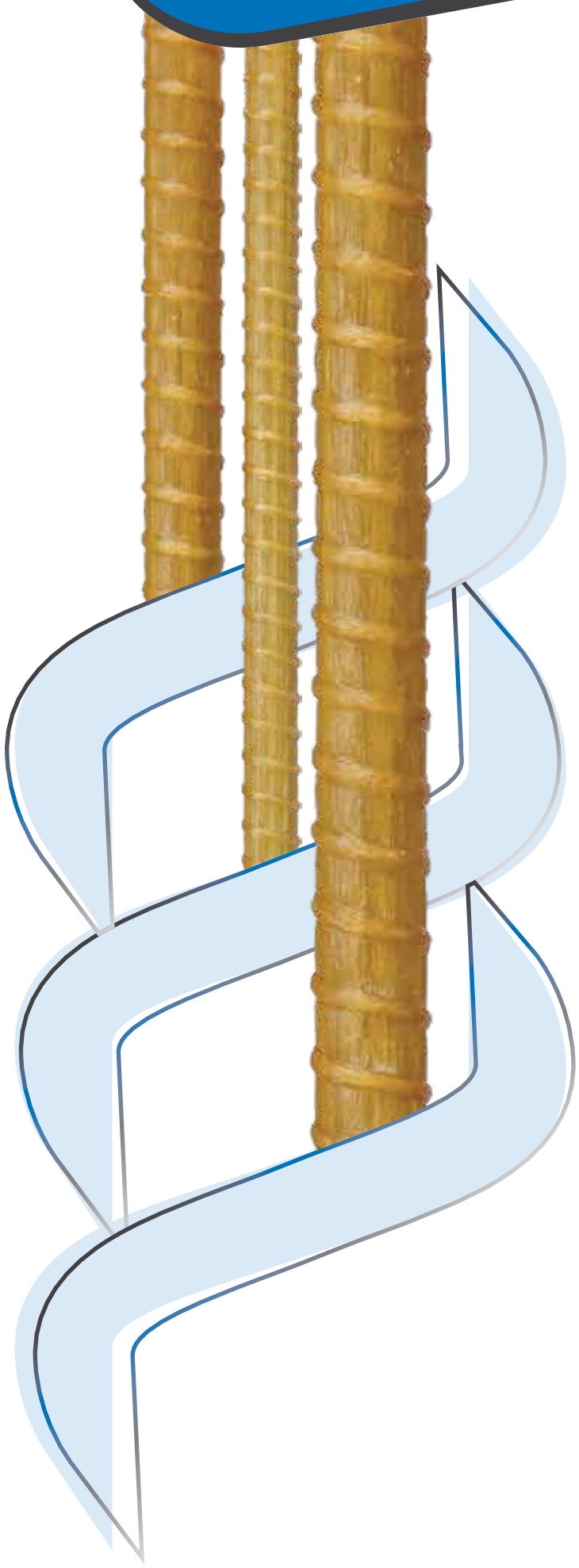
Therefore, the researchers went to the use of FRP composite rebars and produced various models of composite rebars, which left very favorable results to the extent that many structures are made with the help of composite rebars in the world today.

The most commonly used types of composite rebars are:
Carbon Fiber Reinforced Polymer (CFRP), Glass Fiber Reinforced Polymer (GFRP) and Basalt Reinforced Composite Rebar (BFRP). Among them, Glass Fiber Reinforced Polymer (GFRP) is the cheapest and most common type available in the world, which has 3 different resin coatings (epoxy, polyester and vinyl ester resins). The best properties and applications are related to epoxy resins, then vinyl ester and then polyester respectively.

The price of GFRP rebars is much lower than steel rebars and the properties that the user receives from them. Because if the user wants to get the same properties from steel rebar, they have to pay several times more than the price of GFRP rebar.

These rebars significantly prevent the losses caused by the corrosion of steel rebars in corrosive and acidic environments. The use of GFRP rebars in places where there is a need to avoid problems with electric and magnetic fields is the best option for reinforcing concrete.

FRP rebars can also be used to design and build new buildings. With their properties, these reduce the diameter and increase the distance of rebars in reinforced concrete members such as slabs, tunnels, foundations, etc.



Properties of composite rebars

Resistance against Corrosion

Undoubtedly, the most prominent and essential feature of FRP composite products is their resistance against corrosion, which is why they are the best reason to choose them as a substitute for steel rebar. The life of FRP composite rebars is 15 times longer compared to steel rebars due to their very high corrosion resistance.

Electrical and Magnetic Insulation

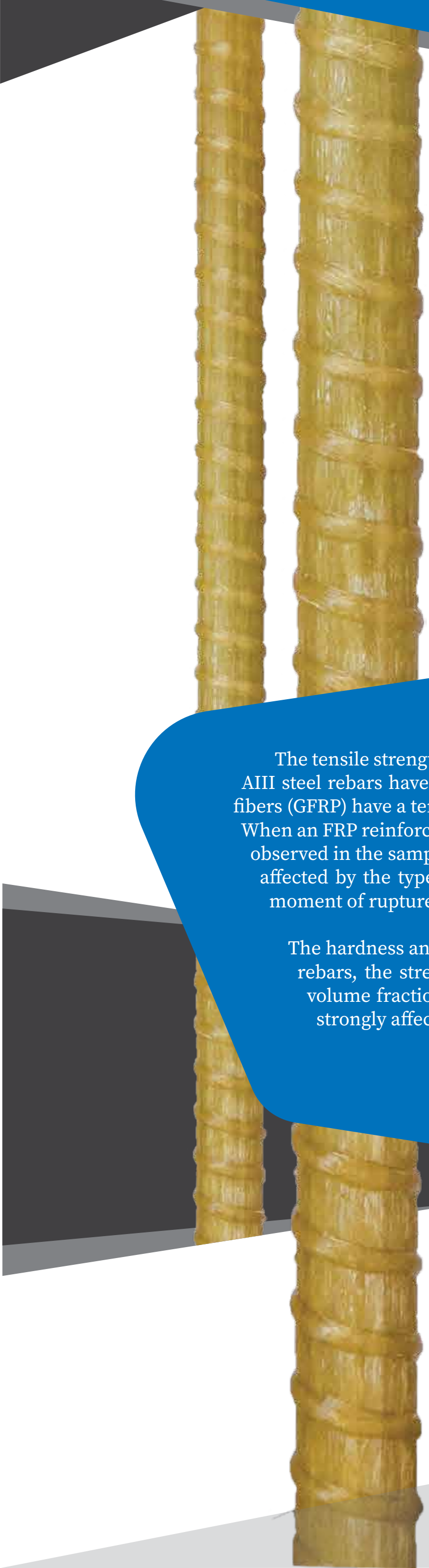
Since these materials are made from a combination of resin and carbon glass fibers or aramid, no stress caused by the magnetic field is introduced into them. Due to this feature in FRP composite materials, it is recommended to use FRP rebars in structures affected by electromagnetic fields such as reactors, airports, MRI departments of hospitals and laboratories, and the foundation of long arc furnaces of steel factories, etc. Electrical conductivity in the structure can be considered a risk. Electrical conductivity in metal materials is a dangerous issue and can cause damage. In contrast to metal materials, FRP composite materials are electrically non-conductive, which makes them a suitable option for structures exposed to electric fields.

Light Weight of FRP Composite Rebars and Significant Reduction in Transportation Costs

Rebars with a specific weight of 2 grams per cubic centimeter are much lighter than common steel bars with a specific weight of 8 grams per cubic centimeter. This reduction in density can lead to reduction in the cost of transportation, easy movement of materials and also reduction in the dead load of the structure.

Resistance against Fatigue

FRP composites are more resistant to alternating loading cycles (fatigue) than steel rebars, so they are widely used in the aerospace industry.



!Economic Savings

The price of this product is between 15 and %50 lower than its steel type.

Tensile Strength

The tensile strength of composite rebars is 3 times higher than the strength of steel rebars. AIII steel rebars have a tensile strength of 480 MPa at best, but composite rebars with glass fibers (GFRP) have a tensile strength of 1600 MPa at best.

When an FRP reinforced composite rebar is subjected to tensile loading, no plastic behavior is observed in the sample until the yield point is reached. The tensile behavior of FRP rebars is affected by the type of fibers and is proportional to their stress-strain behavior until the moment of rupture.

The hardness and tensile strength of FRP rebars depend on several factors. Since in FRP rebars, the strength is basically carried by reinforcing fibers, therefore the type and volume fraction of the fibers used and most importantly the orientation of the fibers strongly affects the tensile properties of FRP rebars.

Application of GFRP Composite rebars



- Sewage facilities such as refineries, clarifiers, manholes, grease traps
- Channels and concrete pipes for directing sewage and industrial effluents and chemicals
- Concrete structures of docks and marine structures
- Structures adjacent to MRI machines in healthcare centers
- Reinforcement of Deck bridges in cold regions and reinforced concrete substructures
- Concrete walls inside subway tunnels
- Nailing and soil strengthening
- Candles in soil and water
- Open channels and concrete pipes for conducting water
- New Jersey prefabricated road concrete blocks
- Prefabricated concrete buildings
- Prefabricated facade panels (GRC)
- Prefabricated walls, columns and prefabricated foundation for wall covering and landscaping
- Power poles and replacement poles of power transmission towers
- Prefabricated concrete power substation room
- Floor reinforcement of parking lots and industrial halls
- Concrete dropped floor and ceiling
- Concrete dams
- Traverses of train tracks
- Foundations
- TBM well
- Ceilings
- Tanks
- Wharf deck and shipping ports
- Retainer walls
- Breakwaters and marine platforms

Mechanical Properties of Composite Ribbed Rebar

MECHANIC PROPERTIES	UNIT	LEVEL	STANDARD TESTING
Tensile strength	N/mm^2	483 to 1600 Mpa	ASTM D7205
Tensile rubber module/	N/mm^2	50000	ASTM D7205
Interfacial shear strength	N/mm^2	400	ASTM D4475
shear strength	N/mm^2	460	ASTM D7617
Average strain at failure	%	2,1	ASTM D7205
the amount of glass fibers	%	75	ASTM D2584
Density	g/cm^3	2	ASTM D792
linear thermal expansion coefficient	10^{-5} / °C	1,8	ASTM E831

Technical Comparison of Composite Ribbed Rebars and the Metal Type

Properties	Composite Ribbed Rebars	Metal Ribbed Rebars
Density	2 g/cm3	8 g/cm3
Weight	Much lighter and ¼ weight of metal type	
Tensile Strength	Much higher tensile strength compared to metal type 483 to 1600 Mpa	483 to 690 Mpa
Electrical and Thermal Conductivity	Not conductive	Conductive
Magnetization	NA	Can be magnetized
Resistance against acids and alkaline and corrosive environments	Resistant	Gradually corrodes and damages concrete
Lifespan	Very high lifespan up to several decades	GetS corroded and destroyed in a much shorter time

Design standards for FRP rebars for concrete reinforcement

Minimum ACI 440 is the most authoritative and prolific consensus standards body involved in publishing design guidelines for the use of FRP bars

ACI 440.11-22 design with GFRP bars was compiled by ACI 440.11-22 under the title "Guide for the design and construction of concrete for structures reinforced with FRP bars.

ACI committee 440, 2022, “Building Code Requirements for Structural Concrete Reinforced with Glass Fiber-Reinforced polymer (GFRP) Bars- Code and Commentary (ACI 440.11-22) , ” American Concrete Institute , Farmington Hills, MI, 260 pp.

ACI 440.1R-15: Guide for the Design and Construction of Structural Concrete Reinforced with Fiber-Reinforced Polymer Bars"ACI Committee 440

ACI 440.8-13: Specification for Carbon and Glass Fiber-Reinforced Polymer (FRP) Materials Made by Wet Layup for External Strengthening of Concrete and Masonry Structures " ACI Committee 440

ACI 440.9R-15: Guide to Accelerated Conditioning Protocols for Durability Assessment of Internal and External Fiber-Reinforced Polymer (FRP) Reinforcement" ACI Committee 440

ACI 440.3R-04 "Guide for Test Methods for Fiber Reinforced Polymers (FRP) for Reinforcing and Strengthening Concrete Structures"

ACI 440.4R-04 "Pre stressing Concrete Structures with FRP Tendons"

AASHTO LRFD Bridge Design Guide

This document published in 2009 stated specifications for concrete decks reinforced with GFRP rebars

csa s806

Canadian designers adopted the application of document S806, Regulations for the Use of FRP Bars in 2006

csas-6

Wide use of GFRP rebars in bridge structures in Canada became possible through this important document

fib 9.3

In Europe, the Fédération Internationale du Beton FIB Group 9.3 has published the technical report "Concrete 40", which contains the "state of the art" of FRP reinforcement in RC structures. Norway and Italy have also published domestic design codes for the use of FRP rebars

Canadian Standards Association (CSA),2019, "Canadian Highway Bridge Design Code (CSA S6-19)," CSA Group , Mississauga, ON, Canada, 1185 pp.

Canadian standards association (CSA),2012, (Reapproved 2021), "Design and Construction of Building Components with Fiber Reinforced Polymer (CSA S806-12, R2021)," CSA Group , Mississauga ,ON, Canada, 187 pp

American Association of State Highway and Transportation Officials,2018, AASHTO LRFD Bridge Design Guide Specifications for GFRP – Reinforced Concrete , Washington, DC

ASTM. 2022. Standard Specification for Solid Round Glass Fiber Reinforced Polymer Bars for Concrete Reinforcement. ASTM D7975-22, West Conshohocken, PA:ASTM.

CSA (Canadian Standard Association). 2019. Specification for fiber-reinforced polymers. CAN. CSA S807-19.Mississauga, ON, Canada: CSA

CNR-DT 204/2006, "Guide for the Design and Construction of Concrete Structures Reinforced with Fiber Reinforced Polymer Bars" DIN-1045-1

ASTM D7205 / D7205M-06, Standard Test Method for Tensile Properties of Fiber Reinforced Polymer Matrix Composite Bars,2011

ASTM D2584, Standard Test Method for Ignition Loss of Cured Reinforced Resins, 2011

BRI, 1995, Guidelines for Structural Design of FRP Reinforced Concrete Building Structures, Building Research Institute, Japan

JSCE, 1997, Recommendation for Design and Construction of Concrete Structures using Continuous Fiber Reinforcing Materials, Concrete Engineering Series 23, Japan Society of Civil Engineers, Tokyo, Japan

ISO 1006-1200 " Fibre-reinforced polymer (FRP) reinforcement of concrete — Test methods — Part 1: FRP bars and grids

Standards and certificates obtained

ISO 14001 : 2015
SHAHKAR FEKR TAVANA CO.
(EPOROD)



ISO 9001 : 2015
SHAHKAR FEKR TAVANA CO.
(EPOROD)



HSE - MS
SHAHKAR FEKR TAVANA CO.
(EPOROD)



CE
SHAHKAR FEKR TAVANA CO.



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TECHNICAL DATA

	Standard Test Methods	UNITS	6 mm	10 mm	13 mm	15 mm	20 mm	22 mm	25 mm	30 mm
Weight		gr	56	157	250	350	600	700	970	1350
specified limit of Tensile force	CSA S807 & ISO 10406	MPa	750	750	650	650	600	550	550	500
Test Result (EPOROD)		MPa	1300	1200	1100	1000	900	830	750	650
Specified limit of Tensile modulus	CSA S807	GPa	40 GPa							
Test Result (EPOROD)		GPa	Between 40 and 50 GPa							
Specified limit of shera capacity	ASTM D7957	MPa	150							
Test Result (EPOROD)		MPa	>150							
Bond Strength	ASTM D7913/D7913M	>8 MPa								
Test Result (EPOROD)		>15 MPa								
Glass Transition Temperature	ASTM D3418	>100 °C								
Test Result (EPOROD)		> 112 °C								
Water Absorption	ASTM D570	< 1%								
Test Result (EPOROD)		0.06 %								
Fiber Content Weight (%)	ASTM D2584	> 70 %								
Test Result (EPOROD)		> 76 %								
Alkali Resistance in High pH(13) Solution without Load at 60°C and 90 Days	ACI 440 Method B6	> 80%								
Test Result (EPOROD)		> 85%								

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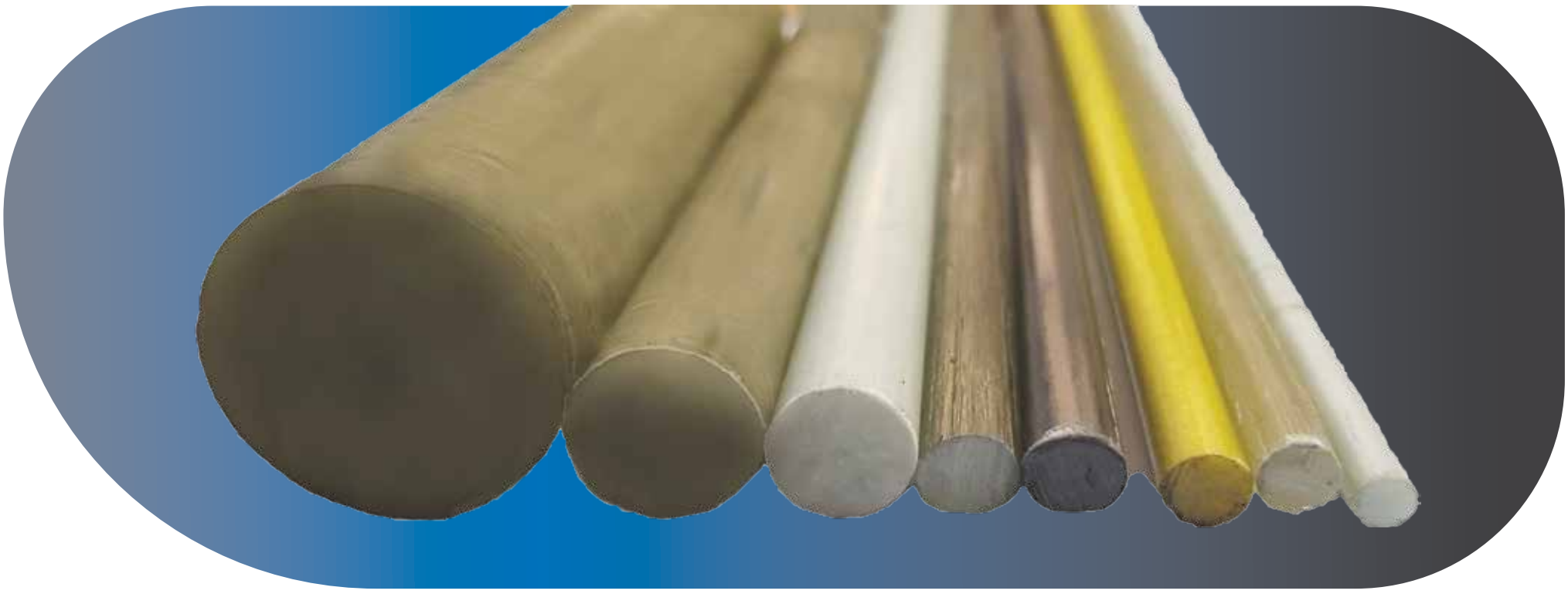
Characteristics and mechanical resistance of composite rebars

FRP Simple Round Composite Rebar or Non-Threaded Rebar

Simple round composite rebar or non-threaded composite rebar is a product with a circular cross-section (without deformations) and a solid end. This product is commonly known as fiberglass rebar, FRP rebar, GRP rebar, GFRP rebar and simple composite rebar. It is possible to produce simple composite rebars (non-threaded) with a diameter of 3 mm and above, in the form of rods from 10 cm to 12 meters or in the form of rolls up to a maximum of 500 meters in different colors.

The advantages of this product include:
Light weight
Corrosion resistance
Chemicals
Electrical insulation
Easy installation and assembly
Threefold tensile strength compared to steel
Stable mechanical performance
High insulation performance with a voltage range of 10KV to 1000KV

The most common method of producing simple composite rebar in the world is pultrusion. Eporod Co., in order to save costs and reduce the price of these products, in addition to the pultrusion production lines has invented a process that combines the pultrusion method (with molding) and non-molding production method. Thanks to this method, the production speed of non-threaded rebars has increased up to 15,000 meters per day.

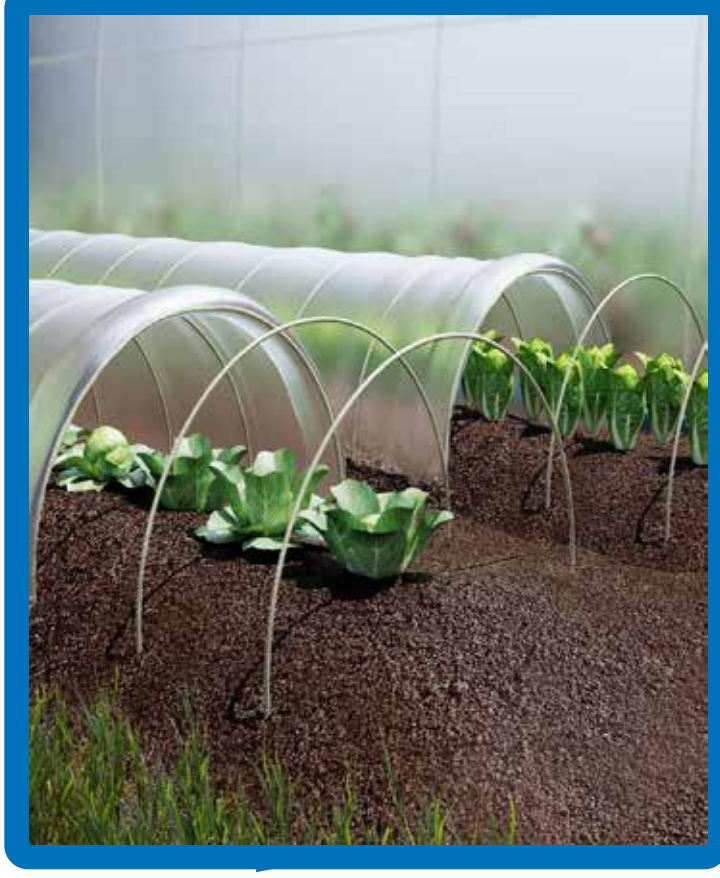


Application of GFRP composite rebars without tread

- Production of composite insulators
- Agricultural support rods
- Greenhouses
- Agricultural nylon support
- Fiber optic cable strength members
- Flag poles and tent poles
- Ladder
- Billboard poles
- Handrail
- Composite fence and stairs
- Bolt and nut
- Handles of tools and sport equipment

Non-threaded rebars produced by Eporod Co. are produced using glass fibers and polyester resin or epoxy resin. They are produced in two grades, anti-UV and regular.





Composite Mesh Networks:

EPOROD proudly announces that for the first time in the world in our beloved country of Iran, it has designed and built a new method composite mesh production line which, through advanced technologies and the company's high capabilities in the field of composites, has been able to manufacture a product of unparalleled quality and thus attract the users' attention.
As a result, by presenting the first modern composite mesh production line, EPOROD has taken an important step towards improving the composite industry in the world, and we hope that by continuing its efforts, it will be able to play a very important role in the composite industry in the future.
This production line was designed and produced by the elite team of EPOROD company using advanced equipment and modern technologies.
Using this production line, EPOROD has been able to produce high quality composite mesh products with high speed in an industrial manner.
By producing composite mesh products in different diameters and dimensions, EPOROD company has been able to introduce itself as one of the top and most popular companies active in the field of composite mesh products for the first time in Iran and by this achievement, increase the reputation and popularity of the company and take another step towards achieving its ambitious goals and the motto of "EPOROD throughout the Middle East"
GFRP (Glass Fiber Reinforced Polymer) composite meshes are used as an alternative to steel meshes in construction. These meshes are composed of a combination of glass fibers and polymer resin and polymer connections and have advantages such as lightness, high flexibility, resistance to breakage and insensitivity to corrosion.

GFRP composite meshes are used in the construction industry to strengthen concrete, improve structural stability, reduce structural weight, and resistance against corrosion. Also, they are used in the construction of complex structures such as shopping centers, bridges, squares, stadiums.
Due to their unique characteristics, GFRP composite meshes can be recognized as a suitable alternative to their steel counterparts in the construction industry.



Advantages of using composite mesh networks and comparison against their steel countertype

- 3 times more tensile strength than steel meshes
- 8 times lighter than steel meshes
- Anti-corrosion and anti-electromagnetism
- Cheaper than steel meshes
- More economical than other methods in terms of saving materials, weight and time
- High resistance due to polymer connections and regular knots
- Speeds up wire placement operation compared to other methods
- Ease of transportation and reduced installation costs
- Optimal use of rebar
- Much smaller number of manpower
- Higher and faster execution speed
- Ready-to-use mesh has more integrity and order in connections and wells.
- Higher resistance due to polymer connections and regularity of the knots.
- Adhesion with concrete is much greater and higher compared to other methods because of polymer joints and the significant thickness and high strength of these joints.
- The anti-corrosion and anti-electromagnetic properties of polymer joints are extremely important in comparison to wire or welding in steel mesh networks in places with high corrosion.



Applications of Rebar Mesh Networks



- All flooring and thermal reinforcements of buildings
- In shotcrete projects
- Precast concrete structures
- Foundation, wall, ceiling and...
- Projects of canal and subway and...
- Composite or concrete walls and prefabricated parts
- The floors of halls, warehouses, highways and airports
- Construction of concrete foundations
- Construction of tunnels, underground and bridges
- Construction of concrete dams, water supply channels
- Construction of concrete silos, concrete tanks and...

No.	Name of Product	Composite Rebar Size	Dimension Well
1	Ribbed and non-ribbed GFRP composite mesh	From 3 mm to 10 mm in diameter	5*5
2	Ribbed and non-ribbed GFRP composite mesh	From 3 mm to 10 mm in diameter	10*10
3	Ribbed and non-ribbed GFRP composite mesh	From 3 mm to 10 mm in diameter	15*15
4	Ribbed and non-ribbed GFRP composite mesh	From 3 mm to 10 mm in diameter	20*20
5	Ribbed and non-ribbed GFRP composite mesh	From 3 mm to 10 mm in diameter	25*25
6	Ribbed and non-ribbed GFRP composite mesh	From 3 mm to 10 mm in diameter	30*30

Composite rebar size: from 3 to 10 mm in diameter
Wells: 5 to 30 cm
Width: 50 to 220 cm
Length: 100 to 1200 cm



Cap or Anchor Head

The composite anchor head is a suitable alternative to the metal bending that is used in many industries. These anchor heads are made of composite materials that include several layers of different fibers that are bonded together with a base resin. These caps have extraordinary features such as high impact resistance, light weight and high durability. Also, they can be used as a protection against electric shock.

Eporod Technical Engineering Company is one of the few companies in the world that has the technology to produce composite caps or anchor heads. Composite anchor heads are also used in petrochemical, construction, marine, and oil and gas facilities.

Considering that each country has its own laws and regulations in the field of production and use of composite products, therefore, in each country there are different regulations and standards for the use of this type of products, and the technical specifications and required tests must also be followed for the production and use of composite caps or anchor heads.

Anchor head or cap is a polymer piece that is designed with ribs. The rebar is inserted through this piece, which has a smaller diameter on one side, and then sinks to the bottom of the anchor head, which has a larger diameter. The outer shape of the anchor head continues as a ribbed design from small diameter to large diameter until it reaches the end of the head, which has the largest cross-sectional area, which is designed for greater binding to concrete.

This ribbed design and the high strength of the piece make the anchor head stick well in concrete and have higher tensile strength than composite and steel bends. Also, this piece has anti-corrosion and anti-electromagnetic properties, which are unique features of this product.
The use of strong and light polymers in the production of anchor head reduces the weight of the part, which helps its easier transportation and installation. Also, the high resistance and anti-corrosion properties of this part can increase its useful life and eliminate the costs of frequent replacement and reconstruction.



Cap or Anchor Head

Using a cap or anchor head instead of bending the ends of steel or composite rebars is a more advanced method with better performance. The cap or anchor head is a polymer material that, unlike metal and composite bends, has a much higher mechanical resistance.

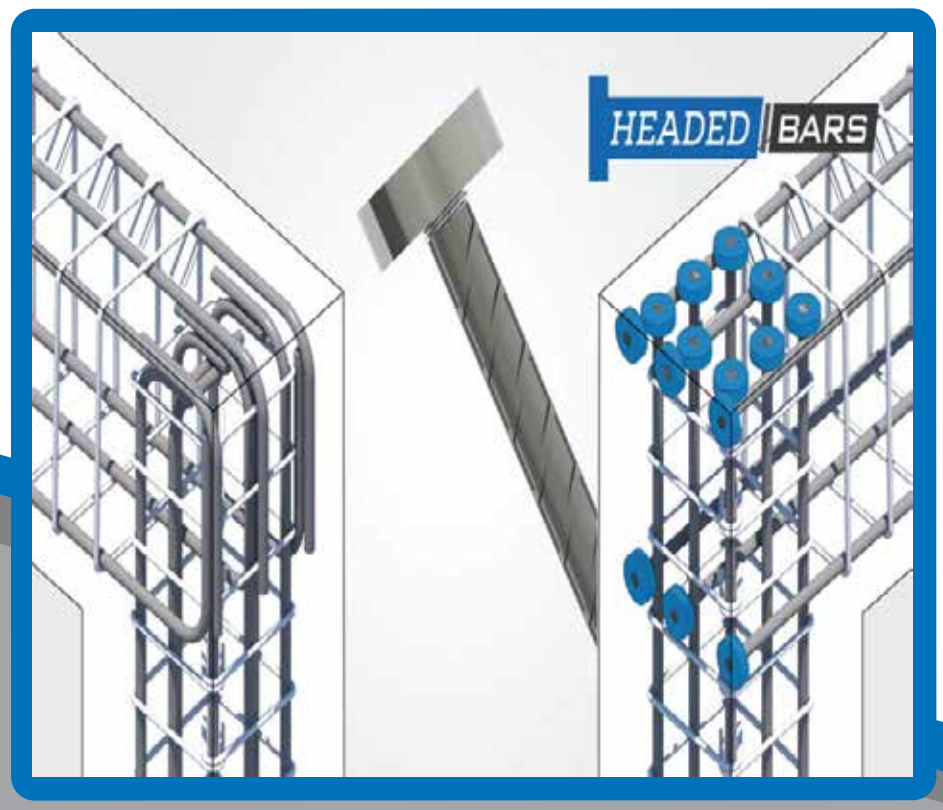
Using a cap or anchor head, the rebars are installed at the point of connection to the concrete. Unlike the bends, these caps connect with the concrete directly and very firmly. Therefore, the adhesion of caps with concrete is much better than that of bends.

Using a cap or anchor head has many advantages. First, the high mechanical resistance of the caps allows to withstand heavy loads and stresses. That is, the ability to transfer axial and anchoring force from rebars to concrete is improved and the possibility of breaking or deformation is minimized.

Moreover, the use of a cap or anchor head is also effective in reducing maintenance and repair costs. Because after some time, if it is necessary to replace the bars, the caps can be simply removed and new rebars installed, without the need to re-bend the rebars and pay the related costs.

In general, using a cap or anchor head instead of bending the ends of the rebars provides better performance in connecting the rebars to the concrete. This method increases mechanical strength and adhesion, reduces costs and increases the useful life of rebars.





Composite Bends, Stirrups and Rebar Bridges

As you know, composite rebars are produced in longitudinal shape due to the use of glass fibers and epoxy resin, which is a thermoset resin.

Unlike steel rebars, which can be bent on site, because composite rebars are made of glass fibers and epoxy resin, and epoxy resin is also thermosetting, it is not possible to bend composite rebars after curing, and if bent or applying too much pressure they will bend to some extent, but after removing the pressure and bending, the rebar will return to its longitudinal shape longitudinal shape due to its elasticity and springiness. Therefore, composite bends, stirrups and rebar bridges are produced according to the listofer provided by the customer in the factory before baking and are sent to the project site. Composite bends, stirrups and rebar bridges are produced in all sizes and dimensions: from 3 mm to 30 mm in diameter and from 20 cm to 60 cm or more in sizes.



Properties

- Anti-corrosion
- More tensile strength than steel
- Faster operation and easier cutting
- Anti-electromagnetic
- 8 times lighter than steel
- Non-conductive against electricity and heat

Composite Bends, Stirrups and Rebar Bridges

The security and useful life of buildings will not be able to be accurately predicted due to environmental problems in humid and coastal areas.One of its factors is the corrosion of steel bars. Corrosion of steel bars causes cracking and flaking of concrete, which will reduce performance and significantly increase reconstruction costs. On the other hand, with the increase of environmental pollution and carbon dioxide and chloride ions, the protective properties of concrete decrease.

The appropriate solution to manage corrosion and reduce the slippage of steel rebar in concrete –both of which cause the concrete to break and honeycomb and as a result the destruction of the structure– is to use composite rebars.

In many projects that are located in humid and corrosive areas, the most important factors include increasing the life of the structure, reducing maintenance and reconstruction costs, further retrofitting, and removing corrosive factors. One of the best and most cost-effective solutions is to replace longitudinal steel bars and steel bends and stirrups with composite bars and bends and stirrups. Due to the anti-corrosion and anti-electromagnetism properties of these polymer products, many problems in corrosive areas can be solved.

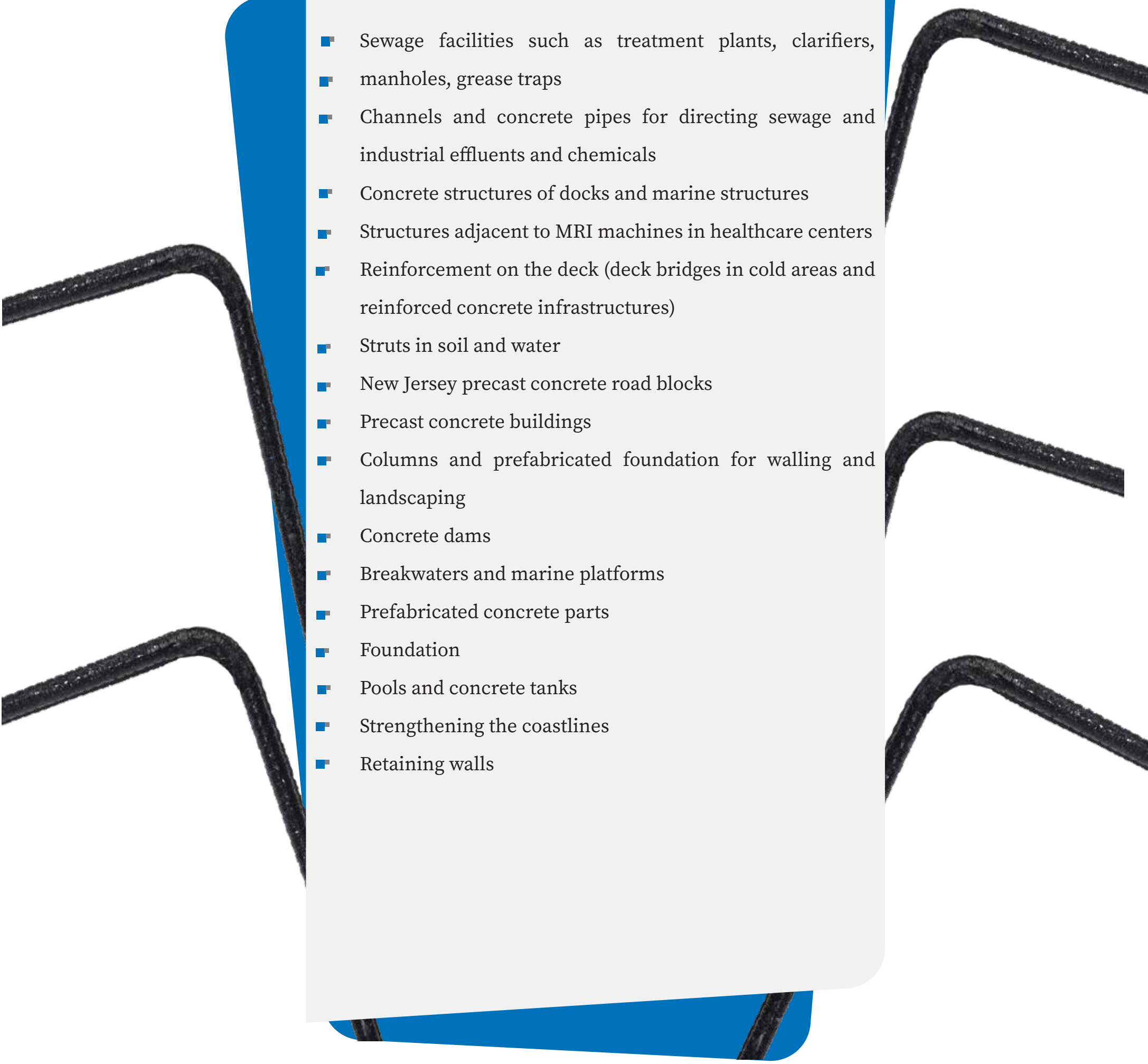
It should be mentioned that composite rebars, which are completely non-conductive and anti-electromagnetic, can be used in projects such as hospital MRI rooms or foundations under electric arc furnaces of steel manufacturing companies or in general any project that requires non-conductivity of the surrounding environment.





Application of Composite Bends and Stirrups

- Sewage facilities such as treatment plants, clarifiers, manholes, grease traps
- Channels and concrete pipes for directing sewage and industrial effluents and chemicals
- Concrete structures of docks and marine structures
- Structures adjacent to MRI machines in healthcare centers
- Reinforcement on the deck (deck bridges in cold areas and reinforced concrete infrastructures)
- Struts in soil and water
- New Jersey precast concrete road blocks
- Precast concrete buildings
- Columns and prefabricated foundation for walling and landscaping
- Concrete dams
- Breakwaters and marine platforms
- Prefabricated concrete parts
- Foundation
- Pools and concrete tanks
- Strengthening the coastlines
- Retaining walls









Precast concrete projects with a length of more than 15 km using GFRP composite rebars produced by Eporod company.





T he project of concrete markers of Gas C ompany (Ministry o f Oil), using GFRP composite rebars pro-d uced by Eporod Company.





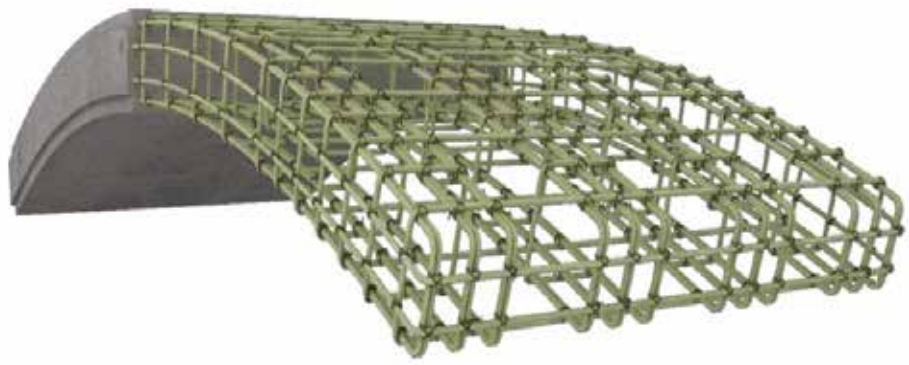


Concrete segments of tunnels using GFRP composite rebars

T he gfrc panels project of I ran's tallest residential tower (Adima), implemented by Inab Neishabour company, using GFRP composi te rebars produced by Eporod company.











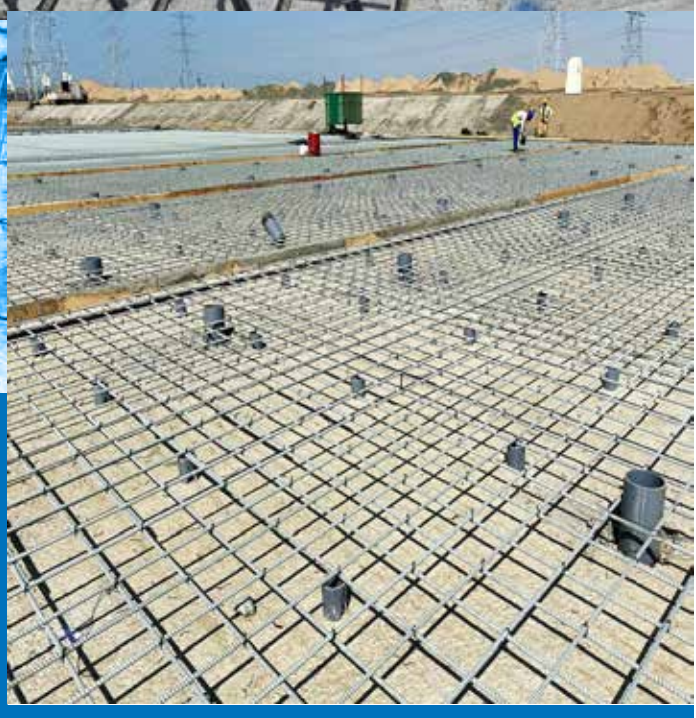
T he large project of the l ong arc furnace founda- t ion of Shadgan Steel Co. u sing GFRP composite rebars of eporod Co

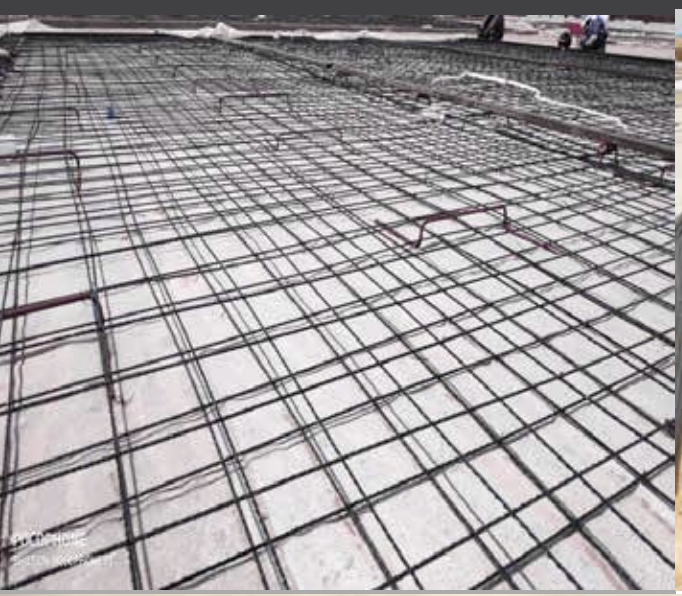






Saudi Jizan water canal pro- ject using GFRP composite rebars











Large floor construction of oil refinery 150,000 with an area of more than square meters using GFRP composite rebars of eporod company











The anti-wear hard flooring project of the helicopter hangar at 17 Shahrivar, Rasht, using GFRP composite bars produced by eporod.





Estakher project, Rasht (implemented byPyazhikEngineer-ing Company), using GFRP composite rebars produced by Eporod company







Gas station floor meshing project for the first time in Iran, using GFRP composite rebars produced by eporod company.







The project of meshing the floor of industrial sheds, using GFRP composite rebars

The project of using GFRP composite rebars for building foundation and radial foundation

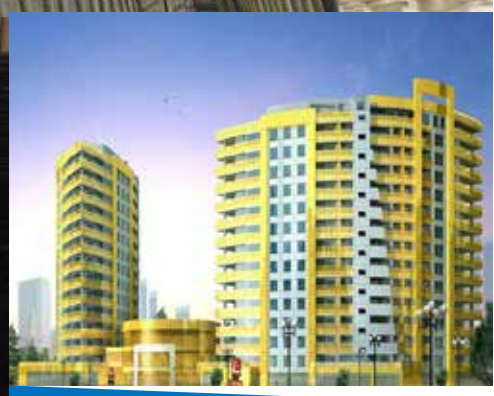






Retaining wall construction project using
From GFRP composite rebars
eporod Co





Parking lot flooring project and
Landscaping of twin towers
Golden Kish





T he project of using GFRP c omposite rebars for the c onstruction of an aquatic breeding pond







Branches and coils of composite rebars

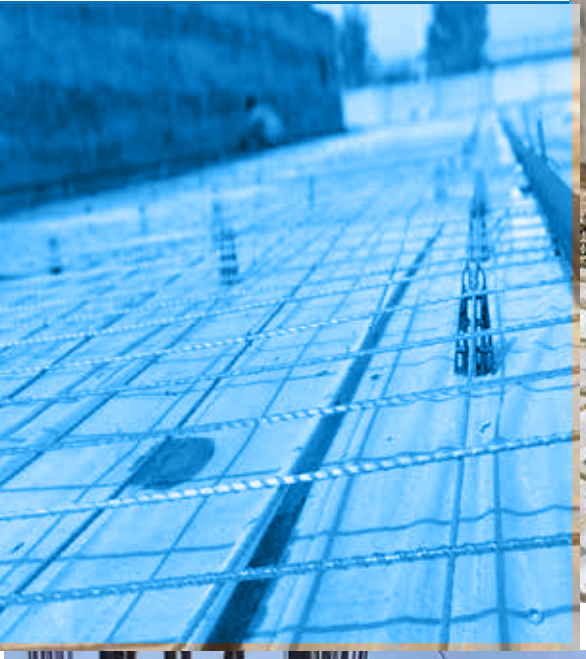
GFRP















Isargaran company's flooring project Using rebars GFRP composite of Eporod company





Bandar Abbas mosque foundation project using GFRP composite rebars of Eporod company





Thermal rebar projects using composite rebars from Eporod company







Types of packaging and transportation of composite rebars

GFRP

















The variety of products produced by Eporod Company















Pictures of Eporod company's equipment and production lines





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